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Synthesis of miktoarm star-shaped polymers with polyoxazoline arms and macrocyclic calix[8]arene branching center

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ABSTRACT

A novel approach to the synthesis of miktoarm star-shaped poly(2-alkyl-2-oxazolines) with calix[8]arene core was developed using a combination of the "core first" (grafting from) and "grafting onto" methods. The star-shaped polymers of A_8B_8 type with grafted poly(2-ethyl-2-oxazoline) and poly(2-isopropyl-2-oxazoline) arms were obtained using calix[8]arene based multifunctional branching center with sulfonyl chloride initiating moieties and as well as acyl hydrazide termination ones. The polymer structure has been confirmed by ^{1}H NMR spectroscopy and UV spectroscopy. The molar mass characteristics of the samples were determined by size-exclusion chromatographic and light scattering. It was experimentally confirmed that all synthesized polymers had the target arm number, namely 8 or 16. The obtained miktoarm stars were characterized by narrow molar mass distributions, and the polymerization degree of the arms was 15. The synthesized stars had a high intra-molecular density, which increased with the arm number. It is shown that the thermoresponsiveness of the studied stars depends on the number and structure of arms and on intramolecular density, while molecular mass is not a decisive factor determining their LCST behavior.

For miktoarm stars, phase separation temperatures do not depend on the grafting configuration of poly(2-ethyl-2-oxazoline) and poly(2-isopropyl-2-oxazoline) on the upper or lower rim of calix[8]arene.

1. Introduction

The star-shaped polymers are a fascinating class of materials that exhibit unique properties due to their structural characteristics. Branched architecture of star-shaped polymers leads to enhanced properties such as improved solubility, increased stability, and higher reactivity compared to linear polymers. In recent years, significant progress was made in the development the synthesis of star-shaped polymers using anionic, cationic and controlled radical polymerization methods. There is currently high interest in star-shaped polymers with a mixed arm structure (so called "miktoarm star polymers") due to their unique structure and versatile applications. These polymers consist of multiple arms of different chemical compositions emanating from a central core, providing a platform for the precise control of material properties. Amphiphilic miktoarm star polymers are showing significant potential in various biomedical applications, such as drug delivery, since they can form nanoparticles of various morphologies as a result of the self-assembly process in selective solvents or under the influence of outer stimulus [1–4]. Furthermore, the stimuli-responsive behavior of miktoarm star polymers allows for triggered drug release in response to external stimuli such as pH, temperature, or light [5,6]. This feature enhances the precision and efficiency of drug delivery systems, making them more effective and targeted.

Although a number of approaches to the synthesis of miktoarm star polymers of various structures have been developed and proposed, their synthesis continues to pose a significant challenge. The complex requirements for orthogonal reactivity, high yields, and specific structure of branching center results in the need of long-term operations, such as coupling and polymerization reactions, protection/deprotection stages, purification steps, including fractional precipitation or high-performance liquid chromatography. Therefore, the challenge of creating novel star-shaped polymers with specific molecular weights and structural features continues to be important.

There are several the most popular methodologies for the synthesis of miktoarm star polymers. One of them is the "core first" (grafting from) strategy with the use of orthogonal functional groups which initiate different types of polymerizations, such as ROP (ring-opening polymerization)/RAFT (reversible addition – fragmentation chain-transfer

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polymerization) [7] or ROP/RAFT/ATRP (atom transfer radical polymerization) [8]. The main limitation of this strategy is that it only allows copolymerization of monomers of different polymerization types. Another commonly used methodology is the "arm first" (grafting through) scheme with a mixed composition of polymer arms in the reaction mixture [9,10]. This strategy allows to synthesize polymers with a large number of arms (spherical polymer brushes), but in this case it is difficult to obtain miktoarm star polymers with a precise structure. To precisely control the structure of miktoarm star polymers, a combination of different schemes is often used, such as "core first/grafting onto" [11,12] or "core first/grafting through" combinations [13]. In fact, these strategies allow forming a miktoarm star polymers from the monomers of a similar type of polymerization mechanism. The combination of the abovementioned schemes with self-assembly processes (host–guest complexation) is also applied to obtain supramolecular miktoarm structures [14,15].

Development of amphiphilic star-shaped structures with the possibility of fine-tuning of their behavior in aqueous solutions continues to be a topical area of research. The synthesis of star-shaped polymers with a mixed composition of polymer arms with different hydrophobic-hydrophilic balance could be the key to solving this problem. In present paper, we propose a novel approach to the synthesis of A_8B_8 type miktoarm star polymers with hydrophilic poly(2-ethyl-2-oxazoline) (PEtOx) and hydrophobic poly(2-isopropyl-2-oxazoline) (PiPrOx) arms. A number of star-shaped poly(2-oxazolines) (PAOx) with various core architectures were obtained [16–18]. However, the synthesis of miktoarm polyoxazolines (N-acylpolyethylenimines) with different alkyl substituents in polymer chains, as well as miktoarm star copolymers with other monomers has not been reported previously. The combination of "core first" and "grafting onto" schemes was applied for the synthesis A_8B_8 type miktoarm star polyoxazolines, whereby the A-type polyoxazoline arms were polymerized on the octafunctional initiator, whereas the B-type polyoxazoline arms were obtained using conventional monofunctional initiator and then were terminated by the acylhydrazide moieties previously introduced in the core structure.

2. Materials and methods

2.1. Materials and reagents

2-Ethyl-2-oxazoline (99 %, Sigma-Aldrich, Germany) was distilled over calcium hydride. 2-Isopropyl-2-oxazoline was synthesized according to the method described in the literature [19] and distilled over calcium hydride. 4-Toluenesulfonic acid methyl ester (methyl tosylate, MeOTs) (98 %, Sigma-Aldrich) was distilled in vacuum. Acetonitrile (99 %, Vekton Reachem, Russia) was distilled over phosphorus pentoxide twice. Hydrazine hydrate (99 %, Vekton Reachem) was used without pre-treatment. Chloroform was distilled over phosphorus pentoxide. Sulfolane was distilled over calcium hydride in vacuum. Multifunctional initiator (branching center) octamethyl tert-butylcalix[8]areneoctaacetate octasulfonyl chloride was synthesized in 3 steps starting from the commercial tert-butylcalix[8]arene (Aldrich) following to the known technique [20]. (^{1}H NMR is shown in Figure S1).

2.2. 2 Characterization of chemicals

^{1}H NMR spectra were obtained on a Bruker AC400 instrument (400 MHz) at 20 °C for solutions in d-chloroform and d6-DMSO. Dialysis procedures were performed using ZelluTrans ROTH dialysis tubes with MWCO 3.5 kDa, 7 kDa and 12–14 kDa. Size-exclusion chromatographic (SEC) analysis of polymer samples was performed on a Shimadzu LC-20 Prominence instrument equipped with Waters Styragel HT4 10 μm (7.8 × 300 mm) column, PSS SDV (8 × 300 mm, 50 Å) column and Shimadzu RID-20A refractive index detector. N,N-Dimethylformamide (99.9 % for HPLC, Aldrich, Germany) with the addition of 5 mM of an anhydrous lithium chloride (LiCl) was used as an eluent at 60 °C temperature and flow rate of 1.0 ml/min.

Solutions of polymer samples with 2 mg/mL concentration, filtered through PFTE hydrophobic syringe filters (ThermoScientific, USA) with 0.45 μm pores, were used for SEC analysis.

2.3. Synthesis of 8-arm star-shaped poly(2-alkyl-2-oxazolines) with calix[8]arene core. General procedure

A vial was loaded with the initiator and monomer at a molar ratio of 1:120, respectively, and sulfolane was added as a solvent. The vial was purged with dry argon and sealed. Reaction mixture was heated at 100 °C for 48 h. After finishing the polymerization process the reaction mixture was diluted with 5 ml of ethanol, and the polymer was purified by dialysis in water using dyalisis tubes with MWCO 7 kDa, then lyophilized.

2.3.1. 8-Arm star-shaped poly(2-ethyl-2-oxazoline)

2-Ethyl-2-oxazoline 2 g (0.02 mol), octamethyl tert-butylcalix[8]areneoctaacetate (C8A) octasulfonyl chloride 0.372 g (0.168 mmol), sulfolane 2 g. Light yellow powder. Yield 1.86 g (93 %).1H NMR spectra of 8-Arm star-shaped poly(2-ethyl-2-oxazoline) is shown in Figure S2 (a) (see Supplementary material).

^{1}H NMR (400 MHz, $CDCl_3$, δ, ppm): 7.6 (m, CH_2), 3.5 (m, $N\underline{CH_2CH_2}N$), 2.2–2.6 (m, $CH_3\underline{CH_2}$), 1.1 (m, $\underline{CH_3}CH_2$).

2.3.2. 8-Arm star-shaped poly(2-isopropyl-2-oxazoline)

2-Isopropyl-2-oxazoline 2 g (0.0156 mol), octamethyl tert-butylcalix[8]areneoctaacetate octasulfonyl chloride 0.393 g (0.178 mmol), sulfolane 2 g. Light yellow powder. Yield 1.92 g (96 %). ^{1}H NMR spectra of 8-Arm star-shaped poly(2-isopropyl-2-oxazoline) is shown in Figure S2 (b).

^{1}H NMR (400 MHz, $CDCl_3$, δ, ppm): 7.6 (m, CH_2), 3.5 (m, $N\underline{CH_2CH_2}N$), 2.6–3.0 (m, $CH_3\underline{CH}CH_3$), 1.1 (d, $\underline{CH_3}CH\underline{CH_3}$).

As a result, two samples of star-shaped eight-armed homopolymer with arms attached along the upper rim of calix[8]arene were synthesized (Fig. 1), namely, poly(2-ethyl-2-oxazoline) (C8A-PEtOx$_{UR}$) and poly(2- isopropyl-2-oxazoline) (C8A-PiPrOx$_{UR}$).

2.3.3. Selective destruction of star-shaped polymers with polymer arms grafted in the upper rim of the calix[8]arene

A sample of 0.1 g of star-shaped poly-2-alkyl-2-oxazoline was mixed

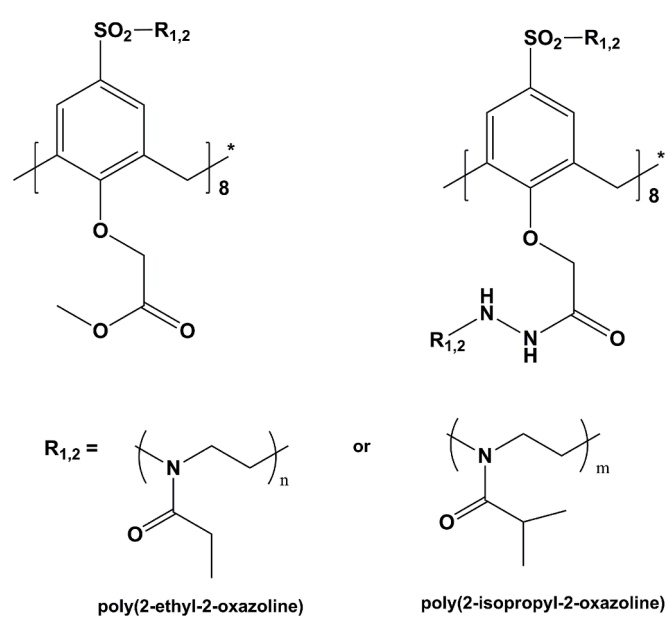

Fig. 1. Structures of star-shaped poly(2-alkyl-2-oxazoline)s.

with an aqueous solution of hydrochloric acid (concentration 1 mol/L). The resulting solution was heated in ampoule at 100 °C for 24 h, then was evaporated in the open cup. The obtained solid residue was dissolved in 5 ml of ethyl alcohol and purified by dialysis in aqueous sodium bicarbonate solution (0.1 mol/L), then lyophilized. The product was dissolved in 15 ml of propionic acid anhydride, heated at 50 °C and stirred for 30 min, then evaporated. To remove traces of moisture, the product was dissolved in anhydrous chloroform and evaporated.

2.4. Introduction of acylhydrazide moieties into the lower rim of the calix[8]arene core of 8-arm star-shaped polyoxazolines

8-Arm star-shaped poly(2-alkyl-2-oxazoline) sample 0.56 g was dissolved in 25 mL of the boiling ethanol. Then hydrazine hydrate 2 mL (0.04 mol) was added gradually and the mixture was allowed to stand for 2 h at ambient temperature. Then solution the resulting material was purified by dialysis in water using dialysis tubes with MWCO 7 kDa, then lyophilized. ^{1}H NMR spectra are shown in Figure S3.

Poly(2-ethyl-2-oxazoline) based chain termination agent. The yield was 0.21 g (38 %).^{1}H NMR (400 MHz, DMSO, δ, ppm): 3.33 (m, $NHNH_2$).

Poly(2-isopropyl-2-oxazoline) based chain termination agent. The yield was 0.49 g (44 %).^{1}H NMR (400 MHz, DMSO, δ, ppm): 3.33 (m, $NHNH_2$).

2.5. Synthesis of linear poly(2-alkyl-2-oxazoline) samples

Linear poly(2-ethyl-2-oxazoline) and poly(2-isopropyl-2-oxazoline) were obtained by the method of living cationic ring-opening polymerization of the corresponding 2-alkyl-2-oxazolines. Toluenesulfonic acid methyl ester (MeOTs) was used as the initiator of polymerization. An ampoule was loaded with the initiator-to-monomer molar ratio of 1:15, respectively, and dry acetonitrile was added. The mixture was heated at 70 °C for 12 h. Samples of linear polymers were taken from the resulting mixtures and evaporated for the subsequent analysis by means of ^{1}H NMR spectroscopy (Figure S4) and SEC (Figure S5).

^{1}H NMR (400 MHz, $CDCl_3$, δ, ppm): 3.5 (m, $N\underline{CH_2CH_2}N$), 2.45 (m, $CH_3\underline{CH_2}$), 1.1 (m, $\underline{CH_3}CH_2$).

^{1}H NMR (400 MHz, $CDCl_3$, δ, ppm): 3.49 (m, $N\underline{CH_2CH_2}N$), 3.05 (br. $\underline{CH_3}N$), 2.62 (m. $CH_3\underline{CH}CH_3$), 1.11 (m, $\underline{CH_3}CH\underline{CH_3}$).

2.6. Synthesis of 16-arm star-shaped poly(2-alkyl-2-oxazolines) with calix[8]arene core

The solution of poly(2-isopropyl-2-oxazoline) based chain termination agent 0.2 g in 0.5 ml of acetonitrile was injected into the ampoule with the resulting linear poly(2-ethyl-2-oxazoline) polymerization mixture, The mixture was shaken for 2 min and heated at 70 °C for 24 h. After the termination process was finished, the resulting product, 16-arm star-shaped mikto-arm polymer with poly(2-isopropyl-2-oxazoline) attached along the upper rim C8A, and poly(2-ethyl-2-oxazoline) attached along the lower rim C8A, (C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$) was purified by the dialysis in aqueous solution of sodium bicarbonate (0.1 % wt.) using dialysis tubes with MWCO 12–14 kDa and then lyophilized. The yield was 0.34 g. Other 16-arm star-shaped samples were obtained in same manner. There were mikto-arm polymer with poly(2-ethyl-2-oxazoline) attached along the upper rim C8A, and poly(2-isopropyl-2-oxazoline) attached along the lower rim C8A, (C8A-PEtOx$_{UR}$-PiPrOx$_{LR}$) 16-arm poly(2-isopropyl-2-oxazoline) (C8A-PiPrOx$_{UR}$-PiPrOx$_{LR}$) and poly(2-ethyl-2-oxazoline) (C8A-PEtOx$_{UR}$-PEtOx$_{LR}$). The structural formulas of the synthesized stars are shown in Fig. 1.

Samples of mikto-arm polymers were analyzed by means of ^{1}H NMR spectroscopy.

^{1}H NMR (400 MHz, $CDCl_3$, δ, ppm): 3.5 (m, $N\underline{CH_2CH_2}N$), 2.45(m, $CH_3\underline{CH_2}$), 1.1 (m, $\underline{CH_3}CH_2$).

^{1}H NMR (400 MHz, $CDCl_3$, δ, ppm): 3.5 (m, $N\underline{CH_2CH_2}N$), 2.62 (m. $CH_3\underline{CH}CH_3$), 1.1 (m, $\underline{CH_3}CH\underline{CH_3}$).

^{1}H NMR (400 MHz, $CDCl_3$, δ, ppm): 3.5 (m, $N\underline{CH_2CH_2}N$), 2.62 (m. $CH_3\underline{CH}CH_3$), 2.45 (m, $CH_3\underline{CH_2}$), 1.1 (m, $\underline{CH_3}CH_2$, $\underline{CH_3}CH\underline{CH_3}$).

^{1}H NMR (400 MHz, $CDCl_3$, δ, ppm): 3.5 (m, $N\underline{CH_2CH_2}N$), 2.62 (m. $CH_3\underline{CH}CH_3$), 2.45 (m, $CH_3\underline{CH_2}$), 1.1 (m, $\underline{CH_3}CH_2$, $\underline{CH_3}CH\underline{CH_3}$).

The polymers are also analyzed by means of ^{13}C NMR spectroscopy. (Figure S6).

^{13}C NMR (400 MHz, $CDCl_3$, δ, ppm): 178 (m, $\underline{C}OCH(CH_3)_2$), 177 (m, $\underline{C}OCH_2CH_3$), 129–96 (br, $arCH_2$), 45(m,$N\underline{CH_2CH_2}N$), 43 (m, $CH_3\underline{CH}CH_3$),38(m, $\underline{CH_3}N$), 35(m, CH_2), 30(m, $\underline{CH_2}CH_3$), 20 (m, $CH_2\underline{CH_3}$, $CH(\underline{CH_3})_2$).

2.7. Molar mass and hydrodynamic characteristics determination

Number-average molecular masses M_n and polydispersity indices $D = M_w/M_n$ of linear poly(2-alkyl-2-oxazolines) and separated arms were determined by SEC method [21]. Weight-average molecular masses M_w and hydrodynamic radii $R_{h\text{-}D}$ of star-shaped samples were determined by the static (SLS) and dynamic (DLS) light-scattering in ethanol solutions (dynamic viscosity η_0 = 1.08 cP, density ρ_0 = 0.789 g·cm^{-3}, and refractive index n_0 = 1.359) at 21 °C. Hydrodynamic radius values $R_{h\text{-}D}$ were calculated using the Einstein-Stokes equation $R_{h-D} = \frac{kT}{6\pi\eta_0 D_0}$, where D_0 – diffusion coefficient, k – Boltzmann constant, T – absolute temperature. The D_0 values were obtained by extrapolating to zero concentration the $D(c)$ values determined at concentration c (Fig. 2).

There was no asymmetry of light scattering, and the values of M_w and the second virial coefficient A_2 were determined using the Debye method. The experiments were performed on a Photocor Complex instrument (Photocor Instruments Inc.), which is equipped with a Photocor DL a diode laser (wavelength λ = 632.8 nm and power 5–30 mW). The instrument was calibrated using benzene (R_V = 2.32·10^{-5} cm^{-1}). The correlation function of the light scattering intensity was obtained using the Photocor-PC2 correlator with the channel number 288 and processed using the DynalS software. The solutions, solvent and calibration fluid were filtered into cells which were dust-free previously by benzene. Millipore filters (Millipore Corporation, USA) with a PTFE membrane with the pore size of 0.20 μm were used. The second virial coefficient A_2 for all studied solutions was positive, therefore, ethanol was a thermodynamically good solvent for the studied polymers. A decrease in A_2 was observed for 16-arm star polymers in comparison with 8-arm one, which is possibly due to the increase in their molar mass (MM).

Refractive index increments dn/dc were determined using

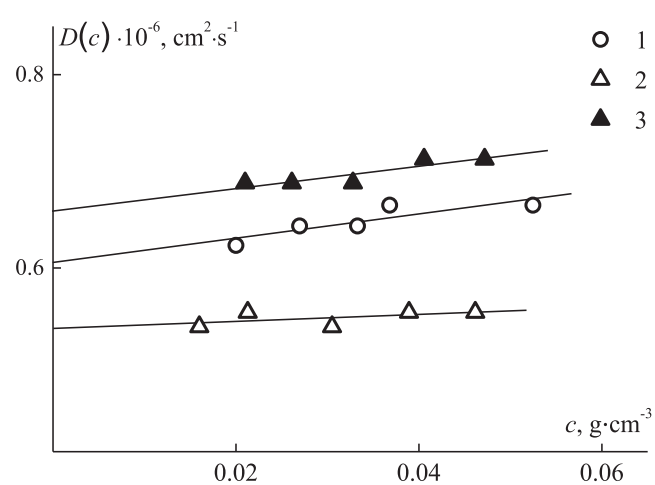

Fig. 2. Concentration dependence of the diffusion coefficients D for ethanolic solutions of C8A-PEtOx$_{UR}$-PEtOx$_{LR}$ (1), C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$ (2) and C8A-PEtOx$_{UR}$ (3).

refractometer RA-620 (KEM, Japan). Intrinsic viscosity was measured with the Ostwald-type Cannon–Manning capillary viscometer (Cannon Instrument Company Inc.) at 21 °C. The solvent efflux time was 50.5 s.

2.8. LCST behavior studies

The thermosresponsive behavior in aqueous solutions of synthesized star-shaped polymers was studied by light scattering and turbidimetry using Photocor Complex instrument. The measurements were carried out in the temperature range T from 15 to 75 °C. The T value was changed discretely with steps from 1 to 5 °C and regulated with an accuracy of 0.1 °C.

The measurement protocol is described in detail previously [22]. For all studied solutions, the values of light scattering intensity I, optical transmittance I^*, hydrodynamic radii R_h of particles presented in solutions, and their contributions S_i to the total light scattering were determined. After establishing a given temperature, the dependences of light scattering intensity I and optical transmission I^* on time t were recorded. These measurements were carried out at the scattering angle of 90°. When I changed less than 1 % during the time required for accumulation and processing of the autocorrelation function, the distribution of I over the hydrodynamic radii R_h was determined and the angular dependences of the values of I, R_h and S_i were analyzed in the angle range from 45° to 135° in order to prove diffusive nature of the modes, and also to obtain extrapolated R_h values (Figure S7).

The concentration of polymers was varied in the range of 0.001–0.01 g·cm^{-3} for C8A-PiPrOx$_{UR}$-PiPrOx$_{LR}$, 0.002–0.02 g·cm^{-3} for C8A-PEtOx$_{UR}$-PiPrOx$_{LR}$ and C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$, 0.005–0.02 g·cm^{-3} for C8A-PEtOx$_{UR}$-PEtOx$_{LR}$. The solutions were filtered into dust-free vials using Millipore syringe filters with a hydrophilic PTFE membrane and a pore size of 0.45 μm (Merck).

3. Results and discussion

3.1. Synthesis of star-shaped PAOx and characterization of their structure

Star-shaped polymers with low polydispersities can be synthesized by various approaches such as "grafting from" or "grafting onto". Additionally, a method that integrates both these approaches is considered, it makes possible to obtain miktoarm star polymers with arms of different nature. The "grafting from"–"grafting onto" scheme combines several polymerization mechanisms, such as ring-opening polymerization and others: RAFT, ATRP or "click chemistry" [23,24,25].One of the key requirements of this approach is the availability of the branching center with an orthogonal set of functional groups, which allows the sequential polymerization of different monomers to create a complex architecture [26].

As it is known, calix[4,8]arenes with sulfonyl halide groups at the upper rim was successfully applied as multifunctional initiators for the cationic ring-opening polymerization of 2-alkyl-2-oxazolines. At the same time, the presence of ester functional groups in the lower rim of the calixarene core can be used for the synthesis of miktoarm star-shaped polymers by the "grafting onto" method, when functional groups with high nucleophilicity are introduced into the lower rim. [27].

Amines possess significant nucleophilicity, which ensures rapid and quantitative addition to the 5-th position of the oxazolinium cation, leading to termination of the polymerization [28]. Carboxylic or sulfonic acid hydrazides have nucleophilicity similar to amines, and can be used to terminate the cationic polymerization of 2-oxazolines so, the multifunctional calix[4,8]arenecarboxylic acid (tetra- / octa-) hydrazides can be applied for the synthesis of star-shaped polymers with a calixarene core using the "grafting onto" approach. Using the model reaction of calix[4]arene tetra(acylhydrazide) with living oligomers of poly(2-ethyl-2-oxazoline), it was shown that such multihydrazides can be utilized as terminating agents for the cationic ring-opening polymerization of oxazolines [29]. It was found that calix[4]arene tetra(acylhydrazide) provides quantitative termination of the polymerization in an aprotic polar solvents and does not cause side processes, such as the hydrazinolysis of amide groups.

The synthesis of two samples of miktoarm A_8B_8 star-shaped polymers with poly(2-ethyl-2-oxazoline) and poly(2-isopropyl-2-oxazoline) arms with a calix[8]arene core was carried out using a combination of the "grafting from" and "grafting onto" approaches. Ring-opening polymerization of 2-alkyl-2-oxazolines on the sulfochlorinated macrocycle was processed on the first stage, followed by the selective conversion of the ester groups of the lower ring into acylhydrazide moieties and further termination of the living oligomeric poly(2-alkyl-2-oxazoline)s on the obtained macromolecular octa(acylhydrazide) on the second stage. As a result, the miktoarm type star macromolecule was obtained with polyoxazoline arms of different structure grafted both to the upper and the lower rims of calix[8]arene macrocycle (Fig. 3).

The introduction of A-type polyoxazoline arms into the upper rim of the calixarene core at the first stage makes it possible to obtain water-soluble thermosensitive 8-arm star polymers, with ester functional groups on the lower rim of the C8A. These ester groups can be converted to the corresponding acylhydrazide moieties by the reaction with hydrazine hydrate. It was shown that mild hydrazinolysis of ester groups at room temperature does not affect amide groups and does not lead to the destruction of polyoxazolines grafted along the upper rim of the C8A core [30]. Polymers with acylhydrazide groups in the lower rim of the C8A macrocycle were used to terminate growing B-type poly(2-alkyl-2-oxazoline) of different structure and finally the A_8B_8 miktoarm structure was obtained. Linear living B-type oligomers of poly(2-ethyl-2-oxazoline) and poly(2-isopropyl-2-oxazoline) were prepared by the cationic polymerization of 2-ethyl-2-oxazoline and 2-isopropyl-2-oxazoline, respectively, using 4-toluenesulfonic acid methyl ester (MeOTs) as an initiator under conditions that ensure the absence of the irreversible termination reaction [31].

The structure of the synthesized miktoarm star-shaped polymers was confirmed by means of ^{1}H NMR spectroscopy and UV–Vis spectroscopy. In the ^{1}H NMR spectra, signals of the side chain (2.62 (m. $CH_3\underline{CH}CH_3$), 2.45 (m, $CH_3\underline{CH_2}$), 1.1 (m, $\underline{CH_3}CH_2$, $\underline{CH_3}CH\underline{CH_3}$)) and broadened signals of the main chain (3.5 (m, $N\underline{CH_2CH_2}N$)) of both poly(2-ethyl-2-oxazoline) and poly(2-isopropyl-2-oxazoline) were observed, which indicates the presence of arms of A-type and B-type polymer chains in the star polymer structure (Fig. 4 and S8). Due to the long relaxation times and the small molar fraction of aromatic units relative to the polyoxazoline chains, the signal intensities of aromatic protons are low, which prevents reliable identification of the presence of C8A macrocycle in the polymer. However, as can be seen from the UV–Vis absorption data (Fig. 5), the spectrum of the resulted star polymer samples contains a characteristic absorption band of benzene rings (λ = 260–290 nm), which confirms the presence of a C8A branching center in the sample.

3.2. Hydrodynamic and molar mass characteristics

Molar masse and hydrodynamic characteristics of investigated polymers are listed in Table 1. It is well known that the possibility of accurate determination of molar masses of branched systems by the SEC using linear standards is limited. However, their molar mass distribution can be measured quite reliable [32]. SEC experiments have shown that the synthesized samples of star-shaped polymers have a monomodal distribution, which indicates the absence of highly defective macromolecules with the incomplete grafting degree of polyoxazoline arms in the sample (Fig. 6 and S9).

The arm number in the star-shaped PAOx was determined from the ratio of the molar masses of the polymer stars (SLS data in Table 1) and polyoxazoline arms M_{arm} (SEC data in Table 2). In the latter case, the calibration curve was obtained using polyethylene oxide standards, which allows to determine the molar mass characteristics of polyoxazolines correctly [33]. Using the obtained values of M_{arm} (Table 2) the polymerization degree N_a and length L_a of arms was calculated

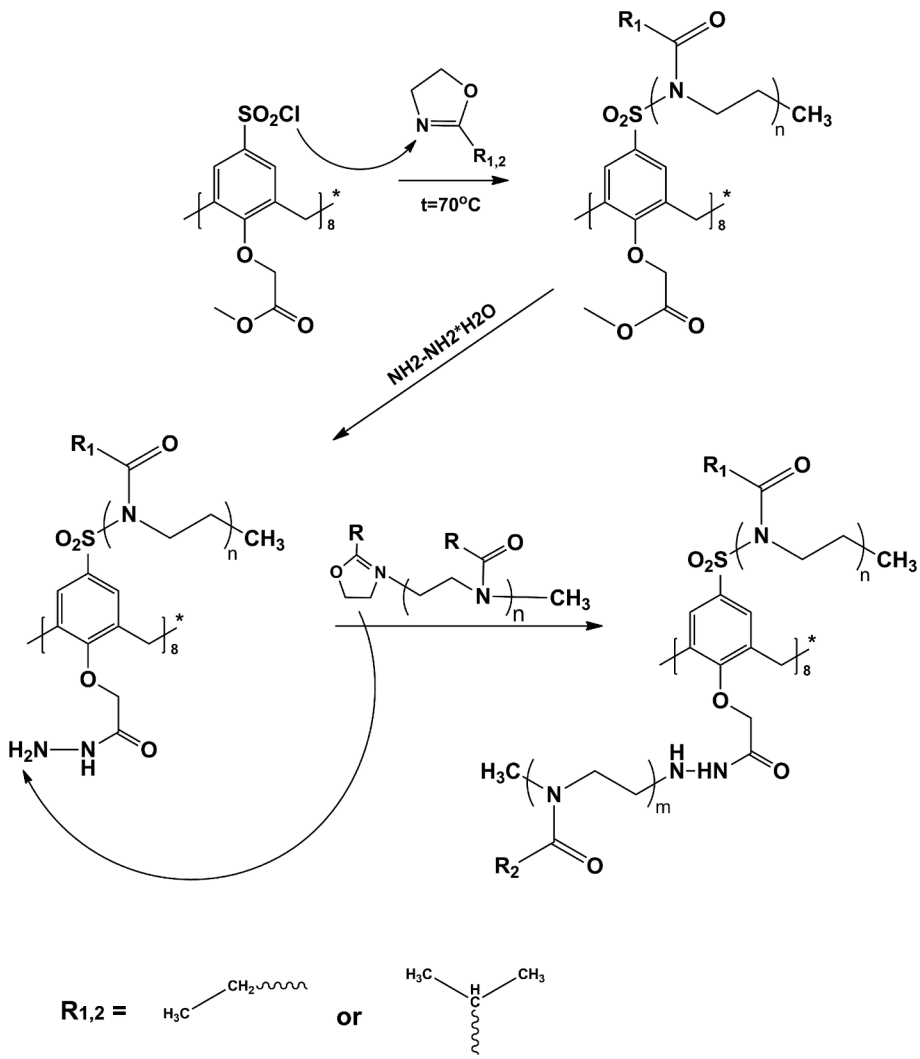

Fig. 3. Reaction scheme for miktoarm A_8B_8 star polymer synthesis using a combined approach.

following the equation (1):

$$L_a = N_a \bullet \lambda_0 \tag{1}$$

where $\lambda_0 = 0.378$ nm is the monomer unit length. The λ_0 parameter was calculated by assuming that all bond angles are tetrahedral and the length of valence bonds is 0.14 nm. The arm number f of star-shaped PAOx was calculated from the ratio of their molecular weights M_{star} obtained by static light scattering and M_{arm}. The arm numbers in the upper f_{UR} and lower f_{LR} rim of the calix[8]arene core was calculated using the equations (2) and (3):

$$f_{UR} = (M_{star8} - M_{C8A})/M_{arm-UR} \tag{2}$$

$$f_{LR} = (M_{star16} - M_{star8})/M_{arm-LR} \tag{3}$$

where M_{star8} и M_{star16} are molar masses of 8-arm star and 16-arm star PAOx, $M_{C8A} = 1928$ g·mol^{-1} is the molar mass of C8A core, $M_{arm\text{-}UR}$ and $M_{arm\text{-}LR}$ are molecular weights of PAOx arms grafted to the upper and lower rim of C8A, accordingly. The total arm number for miktoarm star PAOx was calculated using the equation (4):

$$f_a = f_{UR} + f_{LR} \tag{4}$$

As can be seen from Table 2, for all samples the calculated values of f_a, f_{UR} and f_{LR} coincide with the target number of arms with good accuracy. This allows us to conclude that the proposed combined approach is highly effective for the synthesis of well-defined star-shaped miktoarm poly(2-alkyl-2-oxazoline)s.

The calculated L_a values are close for all star-shaped PAOx (Table 2) and are approximately 1.5 times larger than the values of hydrodynamic radii $R_{h\text{-}D}$ of each sample. This indicates that the arms are folded. At the same time, there is a slight difference in the radii $R_{h\text{-}D}$ of the 8-arm and 16-arm PAOx stars. Accordingly, it can be assumed that the 16-arm PAOx star has a higher intramolecular density. This statement is confirmed by the values of the compression factors, i.e. the ratio of conformational and hydrodynamic parameters of branched polymers compared to linear polymers having the same molar mass. Compression factors decrease with increasing arm number, which represents an increase in the intramolecular density [34]. The viscosity compression factor g^* was calculated using the equation:

$$g^* = [\eta]_{br}/[\eta]_{lin} \tag{5}$$

where $[\eta]_{br}$ and $[\eta]_{lin}$ are intrinsic viscosities of the branched and linear macromolecules, respectively. In order to calculate g^*, it is necessary to know the value of $[\eta]_{lin}$ for linear PAOx. The conformational behavior of poly(2-ethyl-2-oxazoline) has been studied in detail [35,36], while similar information for poly(2-isopropyl-2-oxazoline) is absent in the literature.

Taking into account the slight difference in structure, it can be assumed that PEtOx and PiPrOx have close equilibrium rigidity, then the Mark-Kuhn-Houwink relation obtained for the polymer can be used to

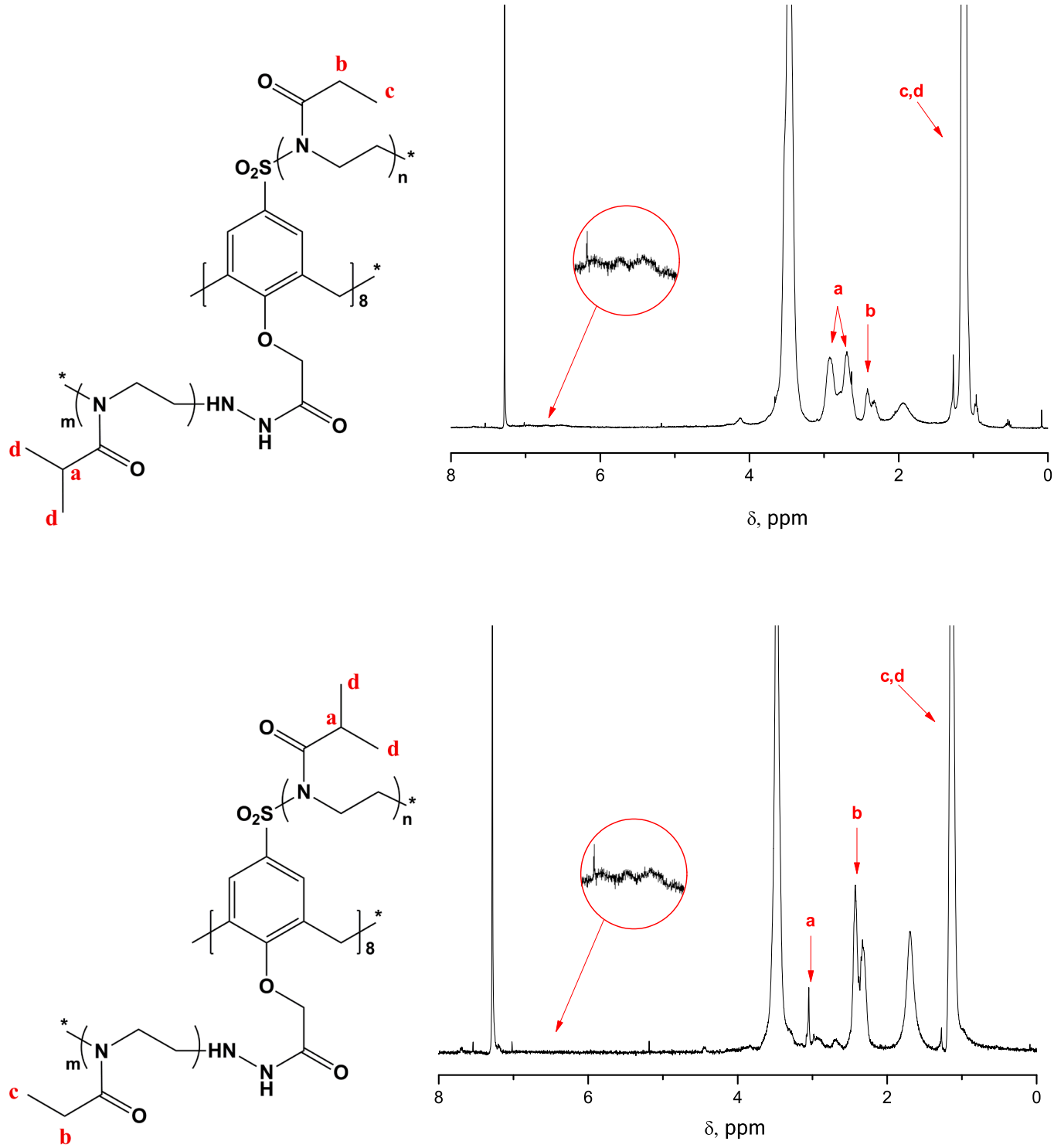

Fig. 4. ^{1}H NMR spectra of 16-arm star polymer with poly(2-ethyl-2-oxazoline) and poly(2-isopropyl-2-oxazoline) arms.

estimate $[\eta]_{lin}$ [36]:

$$[\eta] = 0.045 \bullet M^{0.62} \tag{6}$$

The compression factors obtained using the relationships are shown in Table 1. The g^* values for 8-arm star-shaped PAOx were in the range of 0.35 – 0.37, whereas for 16-arm star-shaped PAOx g^* values significantly decreased to 0.23 – 0.25. This indicates that the fraction of polymeric matter in the volume occupied by the macromolecule is greater for 16-arm stars than for 8-arm ones.

Another confirmation of the increased intramolecular density and compact molecular size of 8- and 16-arm PAOx is the low values of the hydrodynamic invariant A_0, which, in accordance with the definition, is given by the equation [37,38]:

$$A_0 = \eta_0 D_0 (\frac{M[\eta]}{100})^{\frac{1}{3}} / T \tag{7}$$

As can be seen from Table 1, for polymer stars under study $A_0 \leq 2.6 \times$

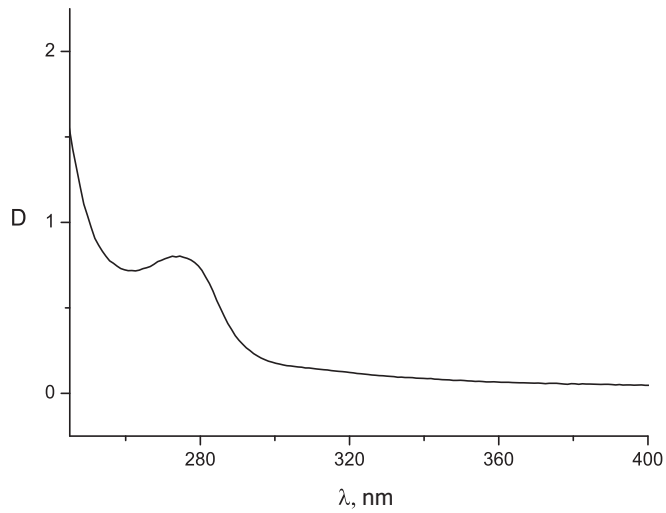

Fig. 5. UV–Vis absorption spectrum of the resulted A_8B_8 star polymer. Solution concentration 0.05% wt.

Table 1
Molar mass and hydrodynamic characteristics of star-shaped PAOx.

Samples	M_w, g·mol^{-1}	$R_{h\text{-}D}$, nm	A_2, 10^{-4} cm^3 mol g^{-2}	[η], cm·g^{-1}	$A_0 \times 10^{10}$, erg·K^{-1} mol$^{-1/3}$	g^*
C8A-PEtOx$_{UR}$	14,500	3.0	8.8	6.3	2.4	0.37
C8A-PEtOx$_{UR}$-PEtOx$_{LR}$	27,500	3.3	4.4	5.9	2.6	0.23
C8A-PEtOx$_{UR}$-PiPrOx$_{LR}$	26,500	3.7	5.5	6.0	2.3	0.24
C8A-PiPrOx$_{UR}$	15,700	3.3	6.8	6.4	2.2	0.36
C8A-PiPrOx$_{UR}$-PiPrOx$_{LR}$	28,100	3.8	4.2	6.0	2.3	0.23
C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$	26,600	3.9	2.2	5.9	2.2	0.24

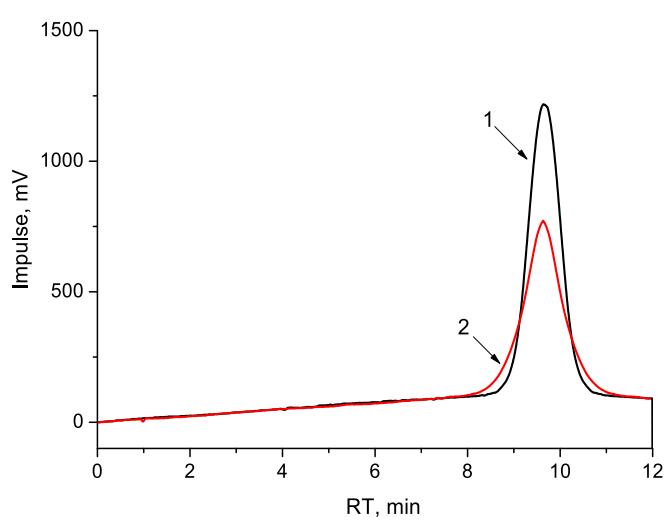

Fig. 6. SEC chromatogram of a 16-arm star polymer C8A-PEtOx$_{UR}$-PEtOx$_{LR}$ (1) and C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$ (2).

Table 2
Structure characteristics of polymer stars under investigation.

Sample	M_{arm}		f_a	f_{UR}	f_{LR}	L_a, nm	
	Upper rim	Lower rim				Upper rim	Lower rim
C8A-PEtOx$_{UR}$	1500	–	8.4	8.4	–	5.7	
C8A-PEtOx$_{UR}$-PEtOx$_{LR}$	1500	1500	17.1	8.4	8.7	5.7	5.7
C8A-PEtOx$_{UR}$-PiPrOx$_{LR}$	1500	1600	15.9	8.4	7.5	5.7	5.3
C8A-PiPrOx$_{UR}$	1700	–	8.1	8.1	–	5.7	–
C8A-PiPrOx$_{UR}$-PiPrOx$_{LR}$	1700	1600	15.9	8.1	7.8	5.7	5.3
C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$	1700	1500	15.4	8.1	7.3	5.7	5.7

10^{10} erg·K^{-1} mol$^{-1/3}$, which is much lower than $A_0 \leq (3.2 - 3.6) \times 10^{10}$ erg·K^{-1} mol$^{-1/3}$ for linear polymers in the coil conformation [37,38]. Low A_0 values were previously observed for star-shaped polymers, molecular brushes, hyperbranched polymers and dendrimers [39–44]. Probably such behavior is typical for polymers of complex architecture, but no theoretical justification has been proposed to date.

3.3. Self-organization of star-shaped PAOx in aqueous solutions

The effect of the arm number on the thermoresponsive behavior for star-shaped PAOx homopolymers was observed even at low temperatures. For example, for 8- and 16-arm PEtOx, a different set of particles was found in their aqueous solutions by the DLS (Fig. 7). For 16-arm star PEtOx the existence of small scattering objects with a hydrodynamic radius R_f and large aggregates with a radius R_s was shown, whereas in aqueous solutions of 8-arm star PEtOx the middle-size particles with a radius R_m and large aggregates were found. On average, the hydrodynamic dimension R_f of small particles was equal to the hydrodynamic radius $R_{h\text{-}D}$ of star-shaped PAOx molecules obtained in ethanol; thus, small scattering objects could be considered as individual star-shaped macromolecules. Their structure is probably similar to that of unimolecular micelles of the core–shell type [45–48], whose arms are strongly folded and shield the hydrophobic core from the solvent. Both

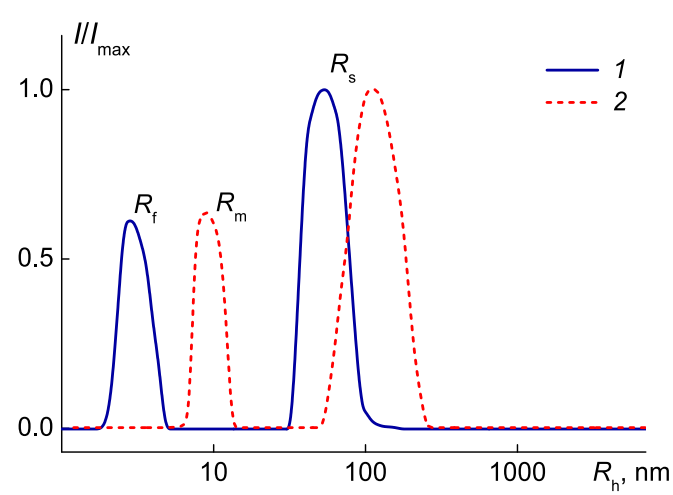

Fig. 7. Dependences of light scattering intensity I/I_{max} on hydrodynamic radii R_h of the scattering objects in the aqueous solutions of C8A-PEtOx$_{UR}$-PEtOx$_{LR}$ (1) and C8A-PEtOx$_{UR}$ (2) (c = 0.005 g·cm^{-3}) at 21 °C.

other particle types were supramolecular structures, which are considered in literature as micelle-like (middle-size particles with a radius R_m) [46,49–52] and loose aggregates (large aggregates with a radius R_s) [49,51,53], respectively. The different behavior of 8- and 16-arm PEtOx can be explained by the higher intramolecular density of 16-arm stars in comparison with to 8-arm stars and, accordingly, better shielding of the hydrophobic C8A core by arms from the solvent. Note that for 16-arm PEtOx the weight fraction of large aggregates in the solution is small. Calculations within the framework of the hard spheres model shown that the weight concentration of these aggregates in aqueous solutions does not exceed 1 %.

The behavior of homopolymeric 8- and 16-arm stars with PiPrOx arms did not differ significantly. At low temperatures, two types of particles existed in aqueous solutions of these samples, namely, micelle-like structures and large aggregates with radii R_m and R_s, respectively. The difference in the behavior of the discussed samples was traced only in the contributions of different particles to the total intensity of light scattering: the concentration of large aggregates was slightly higher for the 8-arm polymer. In aqueous solutions of miktoarm star-shaped PAOx at low temperatures and for all concentrations, two types of scattering objects were detected. These were macromolecules or monomolecular micelles and large aggregates. Moreover, the weight concentration of individual molecules in solutions was much higher than the concentration of aggregates.

The DLS data on the set of scattering objects present in aqueous solutions of PAOx at low temperatures were at least qualitatively confirmed by the AFM (Figure S10). For example, in C8A-PEtOx solutions two types of aggregates with diameters of about 200 and 30 nm were visible. The former correspond to particles with a hydrodynamic radius R_s, and the latter to micelle-like aggregates with a radius R_m. For C8A-PEtOx$_{LR}$-PEtOx$_{UR}$, macromolecules (or unimolecular micelles) and large loose aggregates were recorded.

On heating, LCST behavior was observed for all 16-armed star-shaped PAOx, with the exception of the homopolymer with PEtOx arms (C8A-PEtOx$_{UR}$-PEtOx$_{LR}$). In these cases, for all solutions, regardless of the structure of the arms, one phase transition was detected. A similar situation was observed earlier for star-shaped polymers with block copolymer PAOx arms and the C8A core [54,55]. Concertation dependences of phase separation temperature T_1 for star-shaped PAOx are shown in the Fig. 8.

For investigated stars, the lowest phase transition temperatures T_1 were obtained for the homopolymer 16-arm star with PiPrOx (Fig. 8). Note, that at given concentration, the T_1 values for C8A-PEtOx$_{UR}$-PiPrOx$_{LR}$ solutions are close to phase separation temperatures for 8-arm PiPrOx, with arms grafted at the lower rim [56]. At $c = 0.005$ g·cm^{-3}, the phase transition temperature $T_1 = 38$ °C for the 16-arm PiPrOx is slightly higher than $T_1 = 34$ °C for the 8-arm C8A-PiPrOx$_{UR}$. Typically, for linear polymers, including PAOx, an increase in molar Mass is

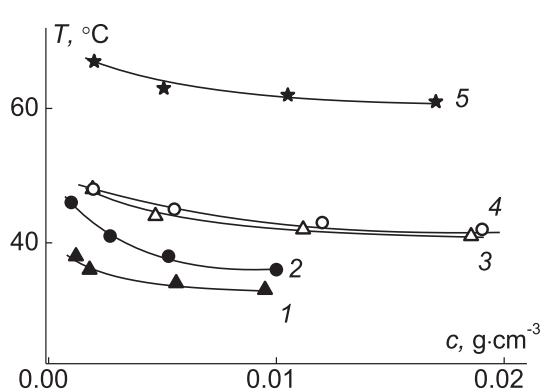

Fig. 8. Concentration dependences of phase separation temperatures T_1 for C8A-PiPrOx$_{UR}$ (1), C8A-PiPrOx$_{UR}$-PiPrOx$_{LR}$ (2), C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$ (3), C8A-PEtOx$_{UR}$- PiPrOx$_{LR}$ (4) and C8A-PEtOx$_{UR}$ (5).

accompanied by a decrease in phase separation temperature and LCST [57]. However, a different situation was observed for the star-shaped polymers studied. Indeed, as noted, a twofold increase in MM for 16-branch PAOx compared to 8-branch PAOx did not lead to a decrease in T_1. These facts can be explained by stronger shielding of the hydrophobic C8A core from the aqueous medium as a result of the higher density of the shell formed by sixteen arms.

Probably, in the same way, the intramolecular density affects the behavior of PEtOx. For aqueous solutions of linear PEtOx oligomers with a polymerization degree $N_a = 15$ (the arms of the studied star-shaped PAOx have approximately the same polymerization degree), no phase separation was detected in the range of 0–100 °C [54]. For the investigated 16-arm star with PEtOx arms, the phase temperature T_1 exceeded 75 °C even at the highest polymer concentration $c \sim 0.05$ g·cm^{-3}. Therefore, despite the star-shaped structure and a sixteen-fold increase in molar mass in comparison with the linear analogue, no LCST behavior was observed for the 16-arm PEtOx.

On the other hand, for a solution of 8-arm C8A-PEtOx$_{UR}$ at $c = 0.005$ g·cm^{-3} the phase transition took place at $T_1 = 63$ °C, and for earlier studied 8-arm star with PEtOx arms grafted to the lower rim of the calix[8]arene also showed LCST behavior ($T_1 = 55.5$ °C at $c = 0.005$ g·cm^{-3}) [55]. That is, a twofold decrease in MM leads not to an increase, but to a significant reduction decrease in temperature T_1. Probably, a decrease in the arm number weakens the shielding of the hydrophobic core from the water and promotes their contacts, thereby increasing the aggregation rate and reducing the phase separation temperature for solutions of 8-armed stars. This is consistent with DLS data obtained at low temperatures. As mentioned above, in solutions of 8-arm C8A-PEtOx$_{UR}$, aggregates existed, while in solutions of 16-arm C8A-PEtOx$_{UR}$-PEtOx$_{LR}$ individual macromolecules were predominated.

For the studied miktoarm 16-arm PAOx, C8A-PEtOx$_{UR}$-PiPrOx$_{LR}$ and C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$, stars in the studied concentration range, close T_1 values were obtained (Fig. 8), which is due to almost the same length of the arms in both samples. In addition, it can be concluded that changing the grafting configuration of PEtOx and PiPrOx arms on the upper and lower rim of C8A does not affect the phase separation temperature, i.e., this temperature is determined by the general hydrophilic-hydrophobic balance of the macromolecules.

On the other hand, for a star in which more hydrophobic PiPrOx arms are grafted to the upper rim of C8A (C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$), an increase in both the hydrodynamic sizes of large aggregates ($R_s = 85$ nm) and their weight concentration in solutions ($c_s = 14$ %) was observed compared to C8A-PEtOx$_{UR}$-PiPrOx$_{LR}$ with the reverse order of arm grafting to calix[8]arene. In C8A-PEtOx$_{UR}$-PiPrOx$_{LR}$ solutions, individual macromolecules predominated, and the aggregates had smaller sizes ($R_s = 55$ nm and $c_s = 1.5$ %). It can be assumed that such self-organization behavior is due to the different conformation of the C8A macrocycle in discussed samples. Indeed, the influence of the calix[8] arene configuration on the processes of self-organization and phase separation was previously established using the example of 8-arm PAOx stars [55,57].

4. Conclusions

A new synthetic technique for the preparation of miktoarm polymer stars with a calix[8]arene core and two different types of polyoxazoline arms has been proposed. The polymers were synthesized by combination of the approaches of “core first” (grafting from) – cationic ring opening polymerization on sulfonyl chloride groups at the upper rim of calixarene – and “grafting onto” – termination of living polyoxazoline chains on hydrazide moieties on the calixarene lower rim. As a result, three types of star polymer samples were synthesized, namely, (i) eight-arm stars with poly(2-ethyl-2-oxazoline) or poly(2-isopropyl-2-oxazoline) arms attached along the upper rim of C8A, (ii) star-shaped 16-arm homopolymers with PEtOx or PiPrOx arms that were grafted onto both the upper and lower rim of C8A, (iii) target micro-arm 16-arm star polymers

in which the grafting geometry of PEtOx or PiPrOx arms was varied.

A comparison of the MMs of 16-arm stars, 8-arm stars and arms showed that the arm number in the synthesized PAOx corresponded to the given one, i.e. eight chains of PEtOx or PiPrOx were grafted onto both the upper and lower rims of the calix[8]arene. The arms of the synthesized stars had almost the same length. Accordingly, the obtained mikto-stars were characterized by narrow molar mass distributions. It has been shown that in a thermodynamically good organic solvent and in water the arms were folded. Therefore, the intramolecular density of the studied star-shaped PAOx was high and increased with the arm number. The influence of the arm structure on the intramolecular density was not detected.

Self-organization of star-shaped PAOx molecules in aqueous solutions also depended on the arm number. This effect manifested itself already at low temperatures, when a change in the number and structure of arms led to a change in the set and size of aggregates formed in the solutions. The higher the f_a value, the denser the outer shell of the molecules formed by PAOx chains, the better the shielding of the hydrophobic core from the water and the lower the aggregation probability.

A dependence of thermoresponsiveness on intramolecular density was found. For example, 8-arm PEtOx demonstrated LCST behavior in contrast to linear PEtOx. This can be explained by the fact that the MM of the stars and the linear analogue differ by 8 times, and also by the fact that the star molecules have a highly hydrophobic branching C8A center. On the other hand, the homopolymer 16-arm star with PEtOx arms did not exhibit thermoresponsiveness, despite the fact that its MM is greater than the MM of 8-arm PEtOx stars. It can be assumed that the reason is the dense hydrophilic corona, which prevents the interaction of the cores and dehydrated PEtOx units of different molecules. For the same reason, the phase transition temperature for the 16-arm PiPrOx was higher than T_1 for the 8-arm PiPrOx. This indicated that MM was not a decisive factor regulating the thermoresponsiveness of investigated star-shaped polymers.

An important parameter which determines the phase transition temperature is the arm structure, therefore the lowest values of T_1 were obtained for solutions of stars with PiPrOx arms. As might be expected, for the miktoarm 16-arm PAOx the phase transition temperature is higher than T_1 for PiPrOx stars and lower than T_1 for 8-arm PEtOx with a C8A core. Most significantly, the grafting configuration of PEtOx and PiPrOx arms on the upper and lower rim of C8A did not affect this temperature, and similar concentration dependences of the phase separation temperature were obtained for C8A-PEtOx$_{UR}$-PiPrOx$_{LR}$ and C8A-PiPrOx$_{UR}$-PEtOx$_{LR}$.

Due to the proximity of the phase transition temperature to the temperature of the human body, the synthesized star-shaped polymers are suitable for use as potential drug delivery systems throughout the body keeping in mind biocompatibility of polyoxazolines. Additionally, they can be used for potential applications, particularly in the removal of pollutants, such as oil from water. Thus, these polymers contribute to an environmentally friendly approach, providing a novel strategy for the synthesis of well-defined miktoarm star polymers following the principles of green chemistry.

CRediT authorship contribution statement

N.D. Kozina: Writing – original draft, Supervision, Investigation. **T. U. Kirila:** Software, Methodology, Formal analysis. **A.N. Blokhin:** Supervision, Methodology. **A.P. Filippov:** Writing – review & editing, Data curation. **A.V. Tenkovtsev:** Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113547.

Data availability

Data will be made available on request.

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